

YUHESEN

FR-mid Ackermann Steering

Drive-by-wire Chassis

User manual V5.0.1



Shenzhen Yuhesen Technology Co., Ltd.

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1. Foreword

(1) Thank you for purchasing our product, this user manual is applicable to FR-mid Ackerman Drive-by-wire Chassis (hereby referred to as "FR-mid").

(2) Before use, please carefully read this user manual and attentions, and correctly use strictly in accordance with this manual.

(3) For the loses caused by serious violation of this user manual, we undertake no responsibilities.

(4) Please well keep this manual for user reference during your operation.

(5) Professionals are required for commissioning, connection and installation of the chassis equipment to avoid irretrievable loses.

(6) DO NOT install, remove or replace equipment lines with electricity. If it is necessary to commission this product with electricity, please select the special commissioning tools with good insulation.

(7) Please use this product under the conditions allowed by laws and regulations, so that the public property or life safety will not be affected.

(8) We will irregularly update this product, the contents of update will be added into the new manual without notification.

(9) This manual may contain the contents which are not correct in technology or which do not comply with the operation. In case of problems which cannot be solved during use of this manual, please contact with the customer service or technical department of us.

(10) As for the contents of this manual, we will try our best to ensure that they are correct and accurate. In case of any improper or incorrect contents, please contact us for confirmation, thank you!

Safety Information

The information herein does not include how to design, install or operate a complete robot, nor the peripheral equipment which may affect the safety of this complete system. The design and use of the complete system comply with the safety requirements formulated in the national standards and specifications. The integrators and end customers of FR-mid are responsible for being sure to comply with practical laws and regulations of relevant countries to ensure that the application of the complete robot will not cause any major danger. These include but are not limited to the following:

■ Effectiveness and responsibilities:

- A risk evaluation shall be conducted to the complete robot system. All the additional safety equipment of other machineries defined by risk evaluation shall be connected. It shall be ensured that, the design and installation of the peripheral equipment of the whole robot system, including software and hardware system, are correct.
- This robot is not equipped with relevant safety functions that a complete autonomously moveable robot shall have, including but not limited to automatic collision avoidance, fall prevention and alarm for creature approaching, etc. For relevant functions, the integrators and end customers are required to conduct safety evaluation in accordance with relevant regulations and feasible laws and regulations to ensure that the developed robot has no any major danger or potential safety hazard during actual application.
- Collecting all the documents of technical files: Including risk evaluation and this manual. Before operation and use of equipment, the existing safety risks may be known.

■ Environments:

- For first use, please carefully read this manual to understand the basic contents and operation specifications.
- For remote operation, please select the areas which are relevantly open. This chassis is not equipped with any sensor for automatic obstacle avoidance.

- This chassis shall be used under the temperature of -20°C~50°C.
- The chassis is not customized for IP protection grade, the IP protection grade of this chassis is IP44.

■ Inspection:

- Inspecting to ensure that the batteries of the equipment are full. Ensuring that the chassis has no abnormality. Inspecting whether the battery of the remote controller is full.

■ Operation:

- Please make sure that the remote control is on when you use it for commissioning, and make sure that the vehicle can receive the remote control commands.
- Ensuring that operation is conducted in a relatively open place. And remote control shall be conducted with sight distance.
- FR-mid The maximum load is 100KG, during use, it shall be ensured that the effective load does not exceed 100KG.
- In case of alarm of low battery of the equipment, please charge timely. In case of equipment abnormality, please stop use immediately to avoid secondary damage.
- In case of equipment abnormality, please contact relevant technicians, DO NOT process without permission.
- Please use the equipment in the environment which meets the IP protection grade requirements of the equipment.
- DO NOT directly push the chassis.
- During charging, please ensure that the environment temperature is higher than 0°C.

■ Maintenance:

- In case of serious tire wearing, please replace timely.
- If the battery will not be used for a long time, when the battery is fully charged, please charge the battery regularly in each month.
- The battery shall be charged once a month at least.

2. Introduction

FR-mid is a versatile drive-by-wire robotics mobile platform, it adopts Ackerman front steering, and rear drive form. Compared with the chassis of differential drive form on the ordinary pavement, FR-mid has a faster traveling ability and relatively strong load capacity. At the same time, the wearing of tire is lighter, matching with whole bridge suspension, the chassis can pass through the common obstacles, such as speed bump, etc. Therefore, it is more applicable for long-term outdoor traveling; And this chassis is a underlayer control system structure based on VCU vehicles control, it uses CAN bus management, having the features of high precision and modularization, etc. By the modules and navigation systems of LiDAR, GPS and manipulator, etc., this chassis is widely used in autonomous driving, unmanned patrol, logistics, transportation distribution, scientific research and various new applications and explorations requiring for mobile chassis.

2.1. Product List

After cargo is delivered, please carefully confirm the product list:

Chassis *1



Remote controller *1



Charger (48V) *1



Product manual *1

YUHESEN

FR_mid 阿克曼线控底盘

用户使用手册

User manual V4.1.0



深圳益森科技有限公司
Shenzhen Yuhesen Technology Co., Ltd.

2.2. Performance Parameters

Table 2-1 FR-mid Parameters

Parameter Type	Performance	Parameter
Structural size and weight	Dimensions(W*D*L)	1320*785*490mm
	Weight	125kg
	Drive	Front Ackermann steering, rear motor drive
	Suspension	Non-independent suspension
	Material	Q235
	Ground clearance	115mm
	Wheelbase	660mm
	Wheel track	665mm
	Tire type/diameter	120/70-10, 420mm
Basic configuration	Driving motor	1000W BLDC motor
	Steering motor	400W servo motor
	Battery type	48V/20AH lithium battery/BMS 48V/40AH lithium battery/BMS(optional)
	Charging time	≤4H
	Charging method	48V/10A manual charging adapter
	External power supply	48V/10A-24V/15A-12V/15A
	Braking mode	Motor brake
	Parking method	Electromagnetic powe-off parking
	Turn signal light	√
	Brake lamp/deceleration indicator/fault indicator	√
	Wheel speed sensor	√
Safety measures	Emergency stop button	√
	Front and rear bumper strip	√
	Command verification	√
	Heartbeat protection	√
	Fault handling for steering system	√

	Fault handling for braking system	√
	Fault handling for driving system	√
	Emergency power off parking protection	√
	Battery fault monitoring and protection	√
	Online detection for whole vehicle CAN node	√
	Whole vehicle fault level division and processing	√
	Vehicle fault warning	√
	Prompt of fast vehicle deceleration	√
	Processing of remote controller disconnection	√
	Charging safety monitoring and protection	√
VCU configuration	Main frequency	168MHz
	Hardware floating point acceleration	√
	Movement control	√
	Communication interface	CAN Interface
	Communication protocol	CAN 2.0B
Performance parameters	Remote control distance	100m
	Vertical load (level road)	100kg
	Speed	23km/h
	Mileage	48V 20Ah battery 30km(full loads)/45km(no loads) 48V 40Ah battery 60km(full loads)/90km(no loads)
	Minimum turning radius	1.7m
	Wading depth	100mm
	Maximum climbing angle	10°(full loads)
	Span width	200mm
	Obstacle surmounting height	60mm(full loads)
	Steering angle	≤0.5°
	IP rating	IP44

	Operating temperature	-20℃~50℃
	Storage temperature	0℃~40℃

3. Product presentation

The contents in this part are only the basic introductions for FR-mid Ackermann Drive-by-wire Chassis, facilitating the users and developers to know FR-mid chassis basically. As shown in Figure 3-1 and Figure 3-2, there are the front and rear overall figure of the whole Ackermann drive-by-wire chassis.

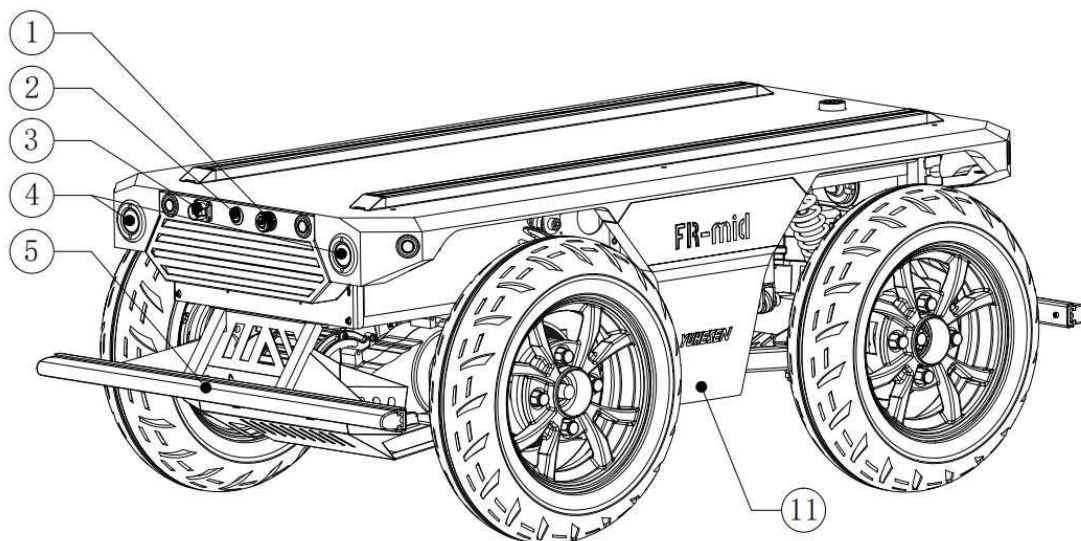


Figure 3-1 rear view

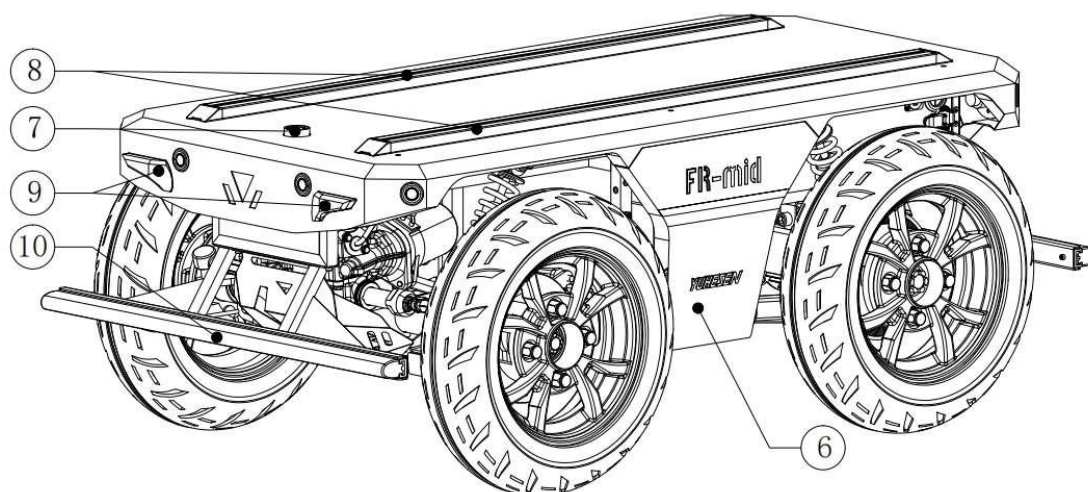


Figure 3-2 front view

Note:

- ① Emergency stop switch;
- ② Power on/off switch;
- ③ Charging port;
- ④ Brake/malfunction lights;
- ⑤ Rear bumper strip;
- ⑥ Right side battery cover;
- ⑦ Top electrical interface;
- ⑧ Standard aluminium mounting rail;
- ⑨ Clearance light / turn signal indicator;

- ⑩ Front bumper strip;
- ⑪ Left side battery cover.

The FR-mid adopts a modular design approach overall, featuring high safety and reliability. Structurally, it employs a front Ackermann steering system and a rear solid axle suspension with a body-on-frame design, ensuring excellent body strength and rigidity, which enhances overall vehicle safety. It boasts strong impact resistance and anti-jolting performance, along with superior off-road capability, enabling it to navigate complex road conditions.

The braking system utilizes an electric motor brake design. Additionally, an emergency stop switch is installed at the rear of the vehicle. When pressed in an emergency, it immediately halts the vehicle, achieving full control. The emergency stop switch also supports functional detection—if it is damaged or disconnected, the VCU prevents the vehicle's drive system from powering up. Multiple safeguards ensure safe vehicle operation.

The chassis features integrated control, with the VCU performing unified analysis and judgment of vehicle signals to form a closed-loop control system. It enables fault diagnosis and timely safety protection measures, reliably achieving remote unmanned vehicle monitoring. The vehicle roof is equipped with open 48V, 24V, and 12V electrical interfaces, as well as communication ports, and comes with standard mounting brackets for rapid secondary development by users.

3.1. State Indicator

Users can determine the vehicle's status through the battery level display and startup sound on the remote control. For details, please refer to Table 3-1.

Status	Description
Battery remaining capacity	The current battery SOC can be viewed by swiping left on the remote control display screen and checking the Value in Ext.V (Figure 3-3). As shown in Figure 3-3, the current battery SOC is 48.62%.

Fault indicator	The vehicle fault status is determined by the flashing frequency of the brake light in non-parking and braking states: 1. flash per second indicates a Level 1 fault alarm; 2. flashes per second indicate a Level 2 fault alarm; 3. flashes per second indicate a Level 3 fault alarm.
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Table 3-1 Chassis status description

Note: Fault level classification and handling methods:

- **Level 1 Fault:** CAN signal and indicator light alarm;
- **Level 2 Fault:** CAN signal, indicator light alarm, and vehicle power reduction;
- **Level 3 Fault:** CAN signal, indicator light alarm, and drive unit shutdown.

Name	ID	Value
TX. V	0	5.55V
Int. V	0	4.96V
Sig. S	0	10
Ext. V	1	48.62V

Figure 3-3 Remote controller vehicle voltage monitoring interface

Note: This interface appears by swiping left on the remote control screen.

- **TX.V:** Current battery voltage of the remote control
- **Int.V:** Power supply voltage of the receiver
- **Sig.S:** Signal strength of the receiver
- **Ext.V:** Battery state of charge (SOC)
- **ID 0:** Signal from the remote control transmitter or receiver
- **ID 1:** First sensor connected to the receiver, and so on.

Voltage(V)	51.03	49.8	49.75	49.74	49.68	49.63	49.52	49.29	49.17	48.97
SOC(%)	100	95	90	85	80	75	70	65	60	55
Voltage(V)	48.96	48.95	48.91	48.82	48.65	48.45	48.19	47.83	47.53	42.65
SOC(%)	50	45	40	35	30	25	20	15	10	7(stop output)

Table 3-2 The Battery voltage vs. SOC reference table

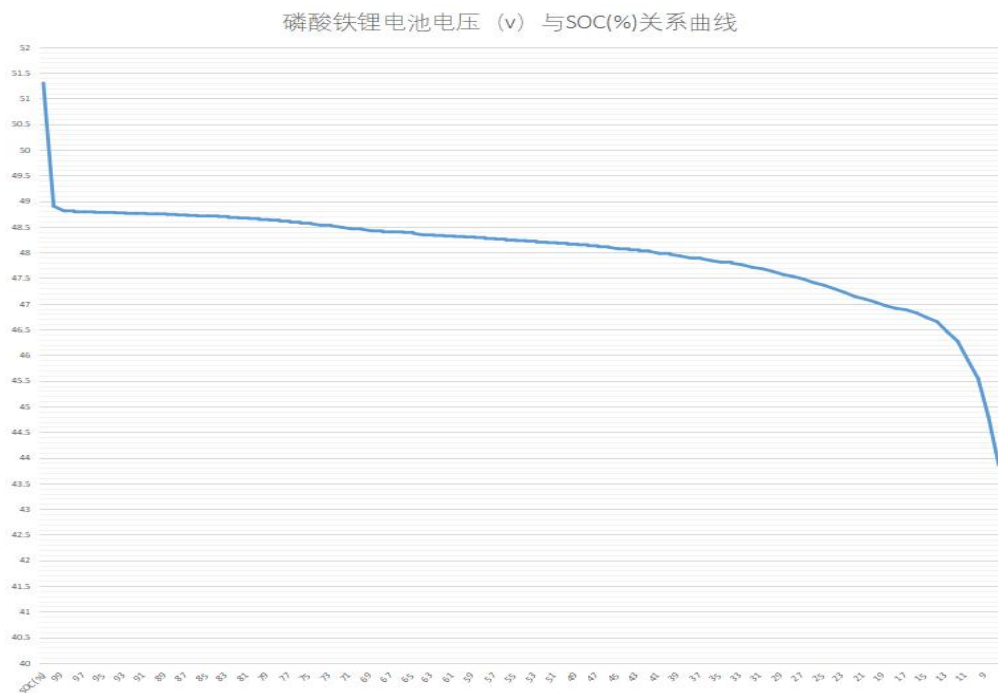


Figure 3-4 The relationship curve between battery voltage and SOC

3.2. Instructions of electrical interface

3.2.1. Instructions of top electrical interface

The FR-mid is equipped with a WS32-12-core aviation electrical connector at the top. This electrical interface is configured with three different power supply groups and one CAN communication interface, with pre-wired leads for easy connection. This allows users to provide power and communication support for various expansion devices.

For the specific location, please refer to the **Top Electrical Interface Layout Diagram (Figure 3-4)**.

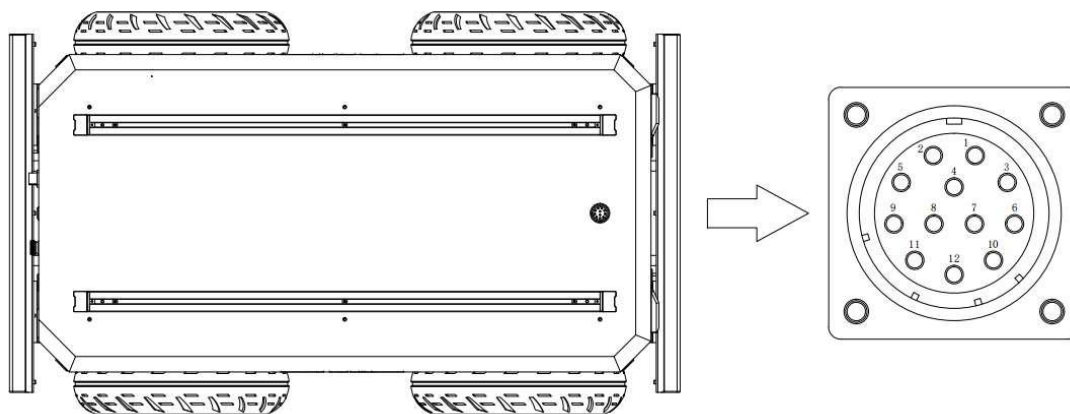


Figure 3-4 Top Electrical Interface Layout Diagram

The specific pin definitions of top electrical interfaces are shown in Table 3-3 below.

Pin	Type	Definitions	Remark
1	CAN	CAN_H	CAN bus
4		CAN_L	
2	BAT	BAT+	Battery 48V power output
3		BAT-	
5	24V	24V-	24V power output
6		24V+	
9	12V	12V-	12V power output
10		12V+	
7	USB	USB_VCC	USB port(reserve)
8		USB_GND	
11		USB_D+	
12		USB_D-	

Table 3-2 Definitions of top electrical interfaces

Please note that the extended power supply here is internally controlled. When the battery voltage is too low, the BMS (Battery Management System) will activate protection and stop battery discharge. Users should pay attention to charging during operation.

3.2.2. Instructions of rear electrical interface

The electrical panel at the rear is shown in **Figure 3-5**, where:

- **B1:** Charging port
- **B2:** Power switch
- **B3:** Emergency stop switch

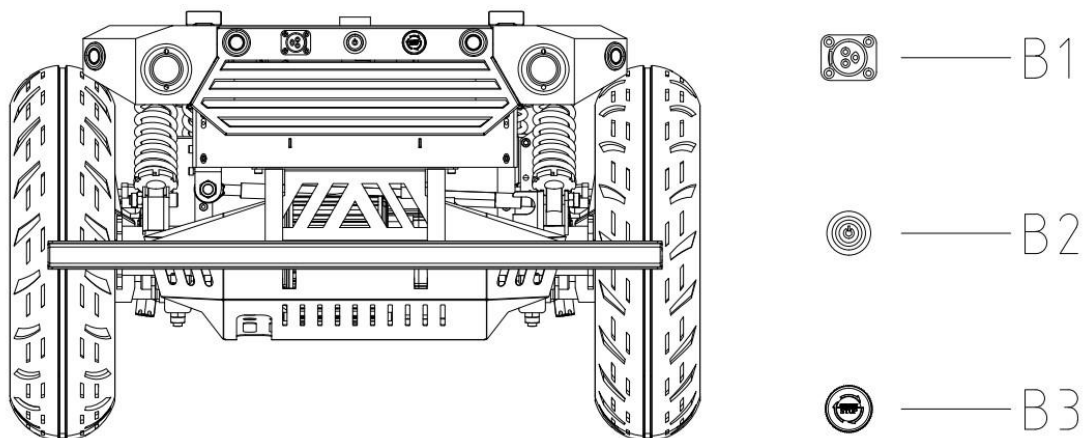


Figure 3-5 Rear electrical panel

3.3. FS-i6S Remote control instructions

The remote controllers have been successfully paired before leaving the factory and require no further adjustments.

Altering the remote controller settings without authorization may cause control confusion, loss of control, and other issues. Please do not modify the settings without proper reason.

If you encounter any parameter configuration problems, please contact our customer service or technical support team.

Should modifications be necessary, they must be performed by qualified technical personnel only.

3.3.1. FS-i6S control instruction

Each FR-mid unit is equipped with an FS-i6S remote controller, allowing users to easily operate the FR-mid.

In this product, the FS-i6S remote controller is configured with the following control scheme:

- **Left hand:** Forward/backward braking
- **Right hand:** Forward/backward throttle and left/right steering

For detailed definitions and functions, please refer to **Figure 3-6**.

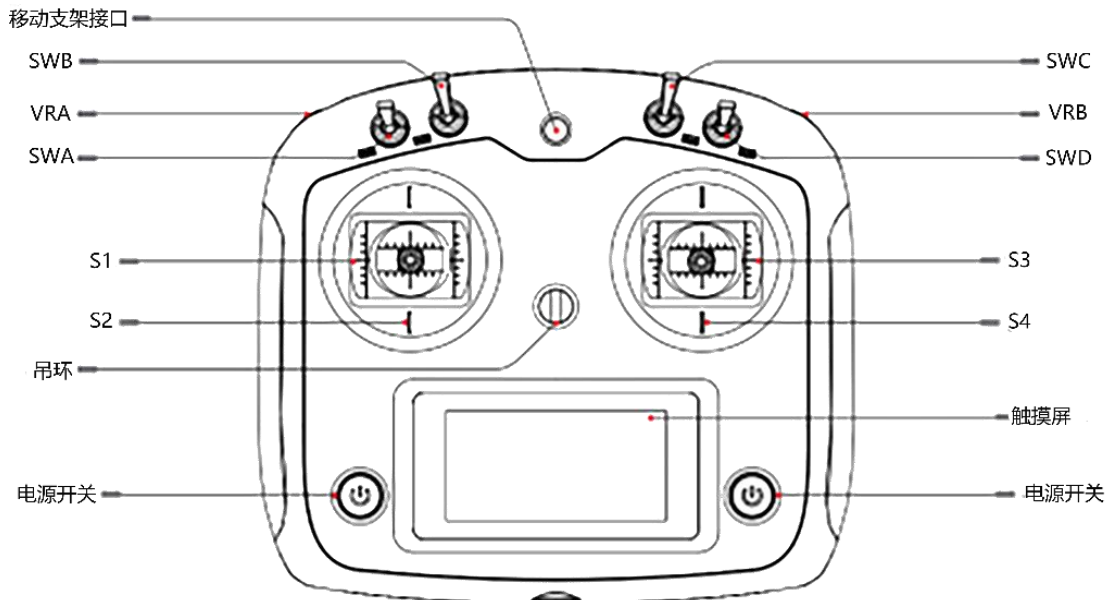


Figure 3-6 FS-i6S Schematic diagram of the FS-i6S remote controller's operation panel

(1) **SWA** is the control mode toggle switch with two positions. With the remote controller facing upward:

- **Upper position:** Remote control mode
- **Lower position:** Command control mode

(2) **SWB** is the gear shift toggle switch with three positions:

- **Middle position (N):** Vehicle is in neutral and does not respond to forward/backward motion signals
- **Upper position (D):** Chassis receives forward motion signals from the **S4** joystick to move forward
- **Lower position (R):** Chassis receives backward motion signals from the **S4** joystick to move backward

(3) **VRB** is the parking request control knob:

- **Upward turn:** Sends a parking request to engage the parking brake
- **Downward turn:** Sends a release request to disengage the parking brake

(4) The **SWD** switch toggles between remote speed and throttle control:

- **Torque control mode:** Enabled if AUTO CAN sends the following commands and SWD is down.
- **Speed control mode:** Enabled if AUTO CAN commands are absent or SWD is up.

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18FFFFFF	0x00	0x00	0x00	0x00	0x00	0x00	0x01	0x00

(5) **S2** is the brake joystick:

- **Push up:** Applies braking force (brake intensity depends on joystick position)
- **Push down fully:** Releases braking force
- **Maximum braking force** is achieved at the top position

(6) Speed & Torque Control Modes

- **Speed Mode (SWD set to Speed):**

SWC position (Up/Middle/Down) sets max speed limit (Low/Medium/High) via **S4** joystick.

- **Torque Mode (SWD set to Torque):**

SWC adjusts acceleration: Economy (Up) / Normal (Middle) / Reserved (Down, defaults to Normal).

(7) Joystick Roles by Mode

- **Torque Mode:**
S4: Throttle level (F/R) | S3: Steering.
- **Speed Mode:**
S4: Speed setpoint (F/R) | S3: Steering.

(8) Power switch: Controls the remote's power state:

- **To turn on:** Long-press both power switches on the display sides when powered off
- **To turn off:** Long-press both switches when powered on
- **If the receiver is active,** the remote cannot be turned off via the switches—remove the battery instead.

Note: The torque control mode is only available when the vehicle is equipped with a hydraulic braking system; otherwise, the vehicle operates solely in speed control mode.

3.3.2. Remote controller alarm instruction

Switch Position Alarm	If the toggle switches (SWA/SWB/SWC/SWD) are not in their default positions during startup, an alarm screen will appear, prompting the user to move all switches to the upward position. The system will proceed to the main interface only when all switches are in their default positions.
Low Voltage Alarm	When the voltage drops below the alarm threshold, the system triggers an alarm, causing the remote controller's screen to flash: TX icon flashes if the remote's voltage is too low. RX icon flashes if the chassis voltage is too low.
Communication Failure Alarm	If the control distance between the remote and chassis is too far or obstructed by environmental interference, the signal strength may drop. A signal strength below 5 triggers a communication failure alarm, warning the user of weak signal conditions.
Remote Controller Inactivity Alarm	If the remote remains unused for an extended period, its buzzer will emit intermittent alarms.

Shutdown Alarm	During shutdown, the remote checks whether the chassis is powered off. If the chassis is still on, a warning screen appears, requiring the chassis to be powered off before the remote can shut down. (To force shutdown the remote while the chassis is still on, remove the remote's battery.)
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Table 3-3 Remote controller alarm status description

3.3.3. Control command and movement instruction

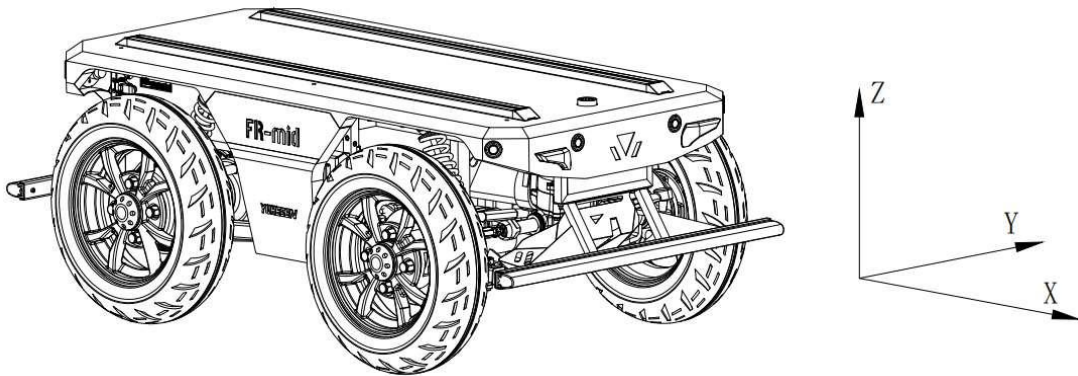


Figure 3-7 Chassis coordinate reference frame

We establish the coordinate reference frame for ground mobile vehicles in accordance with ISO 8855, as shown in Figure 3-7.

As illustrated in Figure 3-7, the FR-mid vehicle body is aligned parallel to the X-axis of the defined reference coordinate system.

Remote Control Mode:

Movement Control:

- Pull down the **VRB knob** to disengage the parking gear.
- Set **SWB toggle** to **D (Drive) gear**—pushing the **S4 joystick forward** moves the vehicle in the **+X direction**.
- Set **SWB toggle** to **R (Reverse) gear**—pulling the **S4 joystick backward** moves the vehicle in the **−X direction**.
- When the **S4 joystick** is pushed fully forward/backward, the speed in the **±X direction** depends on **SWC's high/medium/low speed setting**.

Steering Control:

- The **S3 joystick** controls the front-wheel steering:

- **Push left** → **left turn** (maximum left steering angle at full deflection).
- **Push right** → **right turn** (maximum right steering angle at full deflection).

Torque Mode: Driving & Steering

VRB (hold): Disables parking brake.

- **SWB + S4:**

D-gear + S4 forward: X+ movement (100% at max push).

R-gear + S4 backward: X- movement (100% at max push).

- **S3 Joystick:**

Left/right = steering (max deflection = max wheel angle).

Command Control Mode:

- **Target gear command values:**
- **04:** Movement in the **+X direction**.
- **02:** Movement in the **-X direction**.

Note: The torque control mode is only available when the vehicle is equipped with a hydraulic braking system; otherwise, the vehicle operates solely in speed control mode.

3.3.4. Steering angle control and feedback instruction

Both the vehicle's steering angle control and feedback correspond to the **inner wheel steering angle** during operation:

- **Left turn = Left wheel steering angle** (positive value)
- **Right turn = Right wheel steering angle** (negative value)

As shown in **Figure 3-9 (Ackermann vehicle model)**, the remote controller and CAN control/feedback commands all refer to the α angle.

The equivalent bicycle model steering angle is denoted as β .

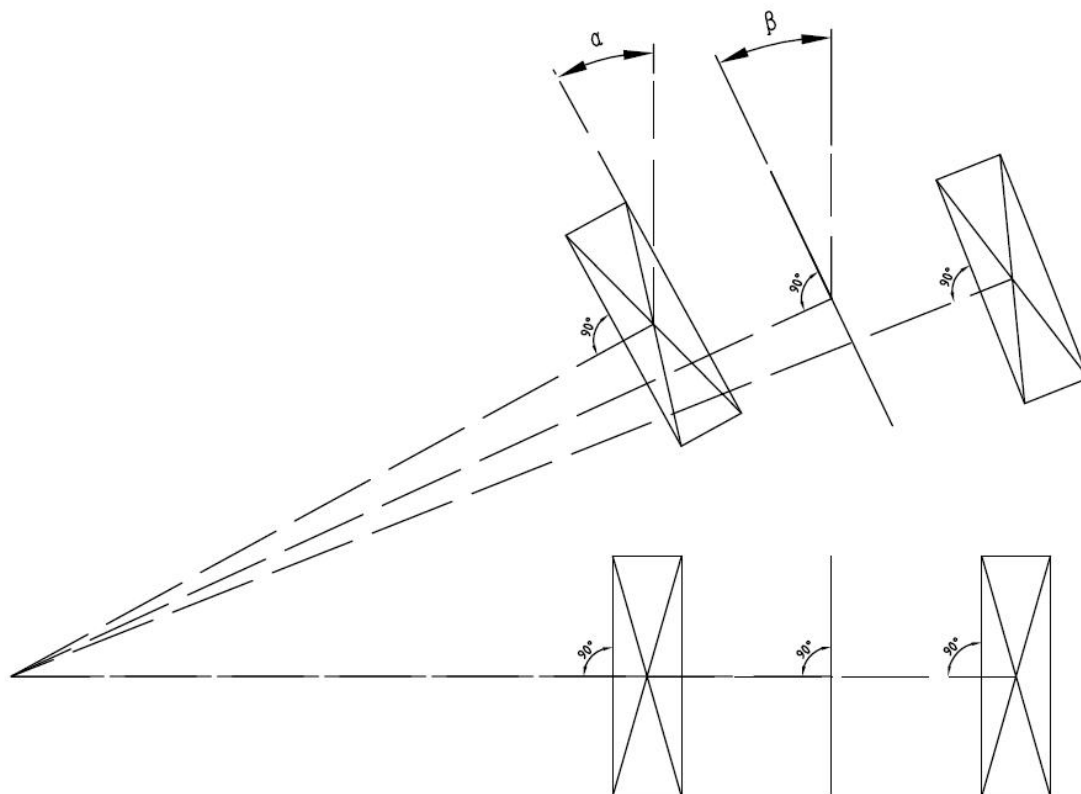


Figure 3-9 Ackermann vehicle model

4. Getting Started

This section primarily introduces the basic operations and usage of the FR-mid platform, detailing how to perform secondary development of the vehicle body through the CAN bus protocol.

4.1. Use and operation

Basic Remote Control Operation Procedure

Inspection

1. **Check the vehicle status.**
 - Inspect for any visible abnormalities. If found, contact after-sales support.
2. **Verify the emergency stop switch status.**
 - Ensure all rear emergency stop switches are in the **released position**.
3. **Confirm all remote control gear positions are in neutral.**

Startup

1. **Power on the remote controller.**
 - Press and hold the power buttons on both sides of the display.
2. **Press B2 (Power Switch).**
3. **Check the remaining battery level on the remote.**
 - If below **30%**, recharge before operation.
4. **Release brakes and parking lock, then switch to remote driving mode.**
 - Observe if the brake lights are flashing, indicating a fault.
 - If a fault is detected, connect a **CAN card** to read error codes and signals, then contact after-sales support for resolution.

Shutdown

- Press **B2 (Power Switch)** again, then release to power off.

Emergency Stop

- Press the **emergency stop switch** on the FR-mid's rear electrical panel.
- The vehicle will enter **emergency stop mode**, with **left & right turn signals flashing** as an alarm.

4.2. Charge

The FR-mid mobile robot chassis comes standard with a **48V/10A charger** to meet users' charging needs.

Charging Procedure:

1. **Before charging**, ensure the FR-mid is **powered off** and confirm that **B2 (Start Switch)** on the rear electrical panel is in the **OFF position**.
2. **Connect the charger:**
 - First, plug the charger's **output connector** into **B3 (charging port)** on the rear electrical panel.
 - Then, insert the charger's **AC plug** into a **220V AC power outlet**.
3. **After charging:**
 - First, **unplug the AC plug**, then **remove the output connector**.
4. **Charger status indicators** are described in **Table 4-1**.

LED indicator status	Charger status
LED1 in bright red	The charger's input cable connector is already energized.
LED2 in bright red	indicate charger is in charging
LED2 in bright green	indicate battery is fully charged

Table 4-1 Charger status indicators description

If the charging environment temperature is too high, the charger may enter temperature protection mode. Please move the charger to a cool or well-ventilated area. Normal charging will resume when the internal temperature drops below 60°C. The charger protection status descriptions are shown in Table 4-2.

Protection Features	Description
Overheat Protection	The charger automatically stops charging when the internal temperature reaches the overheat protection threshold.
Output Short-Circuit Protection	The charger automatically shuts off the output in case of an accidental short circuit.

Reverse Polarity Protection	The charger disconnects the internal circuit from the battery if the battery is connected in reverse.
Output Overvoltage Protection	The charger automatically shuts off the output if an overvoltage condition is detected.

Table 4-2 Charger Protection Status Description

Note:

During vehicle charging, the VCU (Vehicle Control Unit) performs integrated charging protection. If charging is initiated while the vehicle is powered on, the following safety measures are activated:

- Hydraulic braking and electromagnetic parking brake are engaged.
- The motor driver is controlled to shut down high-voltage power for safety.
- Normal operation automatically resumes after charging is disconnected.

Additionally:

- Corresponding CAN signal flags are transmitted to indicate the charging status.
- If necessary, a manual override command can be sent to terminate the charging shutdown process.

4.3. Development

The FR-mid product provides developers with a CAN interface for vehicle control commands, enabling direct communication with the vehicle's electronic systems

4.3.1. CAN protocol

The FR-mid product uses the CAN 2.0B extended frame for communication, with the message format in Intel format and a baud rate of 500K. Through the external CAN bus interface, it is possible to control the chassis movement, including vehicle speed, steering angle, brake pedal opening, and parking requests. The FR-mid will provide real-time feedback on the current motion status information as well as the system status information of the FR-mid chassis.

****Control Operation Description****

(1) ****Speed control mode command frame**** includes gear control, vehicle speed control,

steering angle control, brake opening, parking request, and verification. **Torque control mode command frame** includes gear control, regenerative braking mode, vehicle acceleration mode, accelerator opening, steering angle, brake opening, heartbeat signal, signal verification, etc. For specific protocol details, refer to Section 5.3.4 (for safety reasons, torque control mode is only enabled when the vehicle is equipped with hydraulic brakes).

(2) **Vehicle control priority**: Remote speed mode > Remote torque control mode > Auto speed mode > Auto torque mode > Parking wait mode (for safety reasons, torque control mode is only enabled when the vehicle is equipped with hydraulic brakes).

(3) **Command mode switching**: When the remote control is turned off or the SWA mode switch is in the lower position, and the Remote torque control command frame is online, with heartbeat and verification normal, the system enters torque control mode. Similarly for other control modes: if a higher- priority command frame goes offline, heartbeat stops, or verification fails, the system switches to a lower- priority control mode. If all command frames are offline or CAN signal heartbeat and verification fail, the vehicle stops and enters parking wait mode, awaiting control commands.

****Note:**** To ensure operational safety, the hydraulic braking system is enabled by default when torque control mode is configured; otherwise, torque control mode is disabled.

For wiring instructions, refer to Section 4.3.2; for CAN communication transmission requirements and test examples, refer to Section 4.3.3.

****Caution:**** During operation, it is strictly prohibited to connect CAN wires incorrectly or short- circuit the CAN bus with any type of power supply wire. Failure to comply may result in damage to the electronic control box.

Table 4-3 Command Control Frame and System Feedback Frame

Remote torque control command									
Message Name			ID				Cycle (ms)		Message Length (Byte)
remote_Torque_ctrl_cmd			0x18C4D0D0				10		8
Signal Description	Format	Start ing Byte	Start ing Bit	Sign al Leng th	Data Type	Pre cisi on	Off set	Uni t	Signal Value Description
Target gear	Intel	0	0	4	Unsigned	1	0		00: disable 01: Gear P 02: Gear R 03: Gear N 04: Gear D
Target Acceleration Mode	Intel	0	4	2	Unsigned	1	0		00: normal mode 01: ECO mode 02: Sport mode
Target Energy Recovery Mode	Intel	0	6	2	Unsigned	1	0		00: low energy recovery 01: high energy recovery
Target Vehicle Accelerator Pedal Position	Intel	1	8	8	signed	1	0	%	1%/bit;
Target Vehicle Steering Angle	Intel	2	16	16	signed	0.01	0	°	0.01°/bit;
Target Vehicle Brake Pedal Position	Intel	4	32	8	Unsigned	1	0	%	1%/bit;
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR

									Byte6
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Auto torque control command									
Message Name			ID				Cycle (ms)		Message Length (Byte)
auto_Torque_ctrl_cmd			0x18C4D1D0				10		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Target gear	Intel	0	0	4	Unsigned	1	0		00: disable 01: Gear P 02: Gear R 03: Gear N 04: Gear D
Target Acceleration Mode	Intel	0	4	2	Unsigned	1	0		00: normal mode 01: ECO mode 02: Sport mode
Target Energy Recovery Mode	Intel	0	6	2	Unsigned	1	0		00: low energy recovery 01: high energy recovery
Target Vehicle Accelerator Pedal Position	Intel	1	8	8	signed	1	0	%	1%/bit;
Target Vehicle Steering Angle	Intel	2	16	16	signed	0.01	0	°	0.01°/bit;
Target Vehicle Brake Pedal Position	Intel	4	32	8	Unsigned	1	0	%	1%/bit;
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to

									check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

Auto speed control command									
Message Name			ID				Cycle (ms)		Message Length (Byte)
auto_spd_ctrl_cmd			0x18C4D2D0				10		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Target gear	Intel	0	0	4	Unsigned	1	0		00: disable 01: Gear P 02: Gear R 03: Gear N 04: Gear D
Target vehicle speed	Intel	0	4	16	Unsigned	0.001	0	m/s	0.001m/s/bit;
Target vehicle steering angle	Intel	2	20	16	signed	0.01	0	°	0.01°/bit;
Target vehicle braking	Intel	4	36	8	Unsigned	1	0		
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.

Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6
I/O control command - control frame									
Message Name			ID				Cycle (ms)		Message Length (Byte)
io_cmd			0x18C4D7D0				50		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
I/O control enabled	Intel	0	0	1	Unsigned	1	0		0 = off 1 = on
Steering lamp switch	Intel	1	10	2	Unsigned	1	0		0 = off 1 = left steering lamp on 2 = right steering lamp on
Clearance lamp switch	Intel	1	13	1	Unsigned	1	0		0 = off 1 = on
Charging-forced power-on flag bit	Intel	5	40	1	Unsigned	1	0		Enabling this flag during charging forces the vehicle's 48V high-voltage system to power on, allowing the vehicle to resume driving control. However, when this flag is enabled during charging, the vehicle cannot move in reverse.

Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

Movement control command - feedback frame

Message Name			ID				Cycle (ms)		Message Length (Byte)
ctrl_fb			0x18C4D2EF				10		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Target gear feedback	Intel	0	0	4	Unsigned	1	0		00: disable 01: Gear P 02: Gear R 03: Gear N 04: Gear D
Current vehicle speed feedback	Intel	0	4	16	Unsigned	0.001	0	m/s	0.001m/s/bit;
Current vehicle steering angle feedback	Intel	2	20	16	signed	0.01	0	°	0.01°/bit;
Current vehicle braking status feedback	Intel	4	36	8	Unsigned	1	0		
Current vehicle operating mode feedback	Intel	5	44	4	Unsigned	1	0		0: remote speed control mode 1: remote torque control mode

									2: Auto speed control mode 3: Auto torque control mode 4: Parking mode
Current acceleration mode	Intel	6	48	2	Unsigned	1	0		00: normal 01: eco 02: sport
Current energy recovery mode	Intel	6	50	2	Unsigned	1	0		00: light 01: heavy
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6
Left rear wheel information feedback									
Message Name			ID				Cycle (ms)		Message Length (Byte)
lr_wheel_fb			0x18C4D7EF				10		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Current left rear wheel speed feedback	Intel	0	0	16	signed	0.001	0	m/s	0.001m/s/bit;

Current left rear wheel pulse count feedback	Intel	2	16	32	signed	1	0	1	400 pulses per wheel revolution
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6
Right rear wheel information feedback									
Message Name			ID				Cycle (ms)		Message Length (Byte)
rr_wheel_fb			0x18C4D8EF				10		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Current right rear wheel speed feedback	Intel	0	0	16	signed	0.001	0	m/s	0.001m/s/bit;
Current right rear wheel pulse count feedback	Intel	2	16	32	signed	1	0	1	400 pulses per wheel revolution
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6
I/O control command - feedback frame									
Message Name			ID				Cycle (ms)		Message Length (Byte)
io_fb			0x18C4DAEF				50		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description

				ng th					
I/O control enabling status feedback	Intel	0	0	1	Unsigned	1	0		0 = off 1 = on
Steering lamp switch status feedback	Intel	1	10	2	Unsigned	1	0		0 = off 1 = left steering lamp on 2 = left steering lamp on
Brake lamp switch status feedback	Intel	1	12	1	Unsigned	1	0		0 = off 1 = on
Clearance lamp switch status feedback	Intel	1	13	1	Unsigned	1	0		0 = off 1 = on
Front center bumper strip switch status feedback	Intel	3	25	1	Unsigned	1	0		0 = off 1 = on
Rear center bumper strip switch status feedback	Intel	3	28	1	Unsigned	1	0		0 = off 1 = on
Charging-forced power-on flag bit	Intel	5	40	1	Unsigned	1	0		Enabling this flag during charging forces the vehicle's 48V high-voltage system to power on, allowing the vehicle to resume driving control. However, when this flag is enabled during charging, the vehicle cannot move in reverse.
Charging pile and vehicle contact enable flag bit	Intel	5	41	1	Unsigned	1	0		When the charging station comes into contact with the vehicle in reverse, the VCU charging detection IO port is triggered, and this signal is enabled
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6
Vehicle Running Mileage feedback									

Message Name			ID				Cycle (ms)		Message Length (Byte)
odo_fb			0x18C4DEEF				10		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Mileage	Intel	0	0	32	Unsigned	0.1	0	m	0.1m/bit
Subtotal mileage	Intel	4	32	32	signed	0.1	0	m	0.1m/bit
Battery BMS information feedback									
Message Name			ID				Cycle (ms)		Message Length (Byte)
bms_Infor			0x18C4E1EF				100		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Current battery voltage	Intel	0	0	16	Unsigned	0.01	0	V	0.01V/bit;
Current battery current	Intel	2	16	16	signed	0.01	0	A	0.01A/bit;
Current remaining battery capacity	Intel	4	32	16	Unsigned	0.01	0	Ah	0.01Ah/bit;
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6
Battery BMS flag bit status feedback									
Message Name			ID				Cycle (ms)		Message Length (Byte)
bms_flag_Infor			0x18C4E2EF				100		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description

				th					
Battery SOC	Intel	0	0	8	Unsigned	1	0	%	1%/bit;
Single cell overvoltage protection	Intel	1	8	1	Unsigned	1	0		0 = off 1 = on
Single cell undervoltage protection	Intel	1	9	1	Unsigned	1	0		0 = off 1 = on
Overvoltage protection for the entire battery pack	Intel	1	10	1	Unsigned	1	0		0 = off 1 = on
Undervoltage protection for the entire battery pack	Intel	1	11	1	Unsigned	1	0		0 = off 1 = on
Charging over temperature protection	Intel	1	12	1	Unsigned	1	0		0 = off 1 = on
Charging under temperature protection	Intel	1	13	1	Unsigned	1	0		0 = off 1 = on
Discharging over temperature protection	Intel	1	14	1	Unsigned	1	0		0 = off 1 = on
Discharging under temperature	Intel	1	15	1	Unsigned	1	0		0 = off 1 = on
Charging over current protection	Intel	2	16	1	Unsigned	1	0		0 = off 1 = on
Discharging over current protection	Intel	2	17	1	Unsigned	1	0		0 = off 1 = on
Short circuit protection	Intel	2	18	1	Unsigned	1	0		0 = off 1 = on
Fore-end IC error detection	Intel	2	19	1	Unsigned	1	0		0 = off 1 = on
Software locks up MOS	Intel	2	20	1	Unsigned	1	0		0 = off 1 = on
Charging status	Intel	2	21	2	Unsigned	1	0		0 = not charging 1 = manual charging adapter 2 = front charging station 3 = rear charging station
The current highest temperature of battery	Intel	3	28	12	signed	0.1	0	℃	0.1℃/bit;
The current lowest temperature of battery	Intel	4	40	12	signed	0.1	0	℃	0.1℃/bit;
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and

									disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6
Motor driver encoder signal feedback									
Message Name			ID				Cycle (ms)		Message Length (Byte)
Drive_fb_MCUEncoder			0x18C4DCEF				10		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Driving motor encoder pulse feedback	Intel	0	0	32	signed	1	0		The motor feedback provides 4096 pulses per revolution, with the vehicle's default reduction ratio set at 10:1.
Driving motor torque feedback	Intel	4	32	16	signed	0.01	0		
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

Ultrasonic radar 1 - feedback frame(reserved)									
Message Name			ID				Cycle (ms)		Message Length (Byte)
ultrasonic_1_fb			0x18C4E8EF				100		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
No.1 ultrasonic radar probe's distance	Intel	0	0	12	Unsigned	1	0	mm	This frame signal exists when ultrasonic radar is connected; otherwise, it remains reserved.
No.2 ultrasonic radar probe's distance	Intel	1	12	12	Unsigned	1	0	mm	
No.3 ultrasonic radar probe's distance	Intel	3	24	12	Unsigned	1	0	mm	
No.4 ultrasonic radar probe's distance	Intel	4	36	12	Unsigned	1	0	mm	
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

Ultrasonic radar 2 - feedback frame(reserved)									
Message Name			ID				Cycle (ms)		Message Length (Byte)
ultrasonic_2_fb			0x18C4E9EF				100		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
No.5 ultrasonic radar probe's distance	Intel	0	0	12	Unsigned	1	0	mm	This frame signal exists when ultrasonic radar is
No.6 ultrasonic radar probe's distance	Intel	1	12	12	Unsigned	1	0	mm	

No.7 ultrasonic radar probe's distance	Intel	3	24	12	Unsigned	1	0	mm	connected; otherwise, it remains reserved.
No.8 ultrasonic radar probe's distance	Intel	4	36	12	Unsigned	1	0	mm	
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0		For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0		Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

ECBOX 48V current and voltage feedback									
Message Name			ID				Cycle (ms)		Message Length (Byte)
ebox_48V_voltage_current_st			0x18FFFFFFC				50		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Voltage of 48V Motor driver collecting	Intel	0	0	10	Unsigned	0.1	0	V	
Current of 48V motor driver collecting	Intel	1	10	12	Signed	0.1	0	A	
Voltage of 48V electrical control box collecting	Intel	2	22	10	Unsigned	0.1	0	V	
Current of 48V electrical control box	Intel	4	32	12	Signed	0.1	0	A	

collecting									
Voltage of 48V power output collecting	Intel	5	44	10	Unsigned	0.1	0	V	
Current of 48V power output collecting	Intel	6	54	12	Signed	0.1	0	A	

ECBOX power output current, voltage and ECBOX temperature feedback									
Message Name			ID				Cycle (ms)		Message Length (Byte)
ebox_cus_temp_st			0x18FFFFFFD				50		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Voltage of 24V power output collecting	Intel	0	0	8	Unsigned	0.1	0	V	
Current of 24V power output collecting	Intel	1	8	10	Signed	0.1	0	A	
Voltage of 12V power output collecting	Intel	2	18	8	Unsigned	0.1	0	V	
Current of 12V power output collecting	Intel	3	26	10	Signed	0.1	0	A	
Battery input voltage collecting	Intel	4	36	10	Unsigned	0.1	0	V	
MOS temperature of electrical control box	Intel	6	48	8	Unsigned	1	-50	°C	
Charging MOS temperature	Intel	7	56	8	Unsigned	1	-50	°C	

ECBOX fault status feedback									
Message Name			ID				Cycle (ms)		Message Length (Byte)
diag_ebox_st_fb			0x18FFFFFFB				50		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Charging high temperature limit	Intel	0	0	1	Unsigned	1	0		0 = normal 1 = faulty
Charging high temperature shutdown	Intel	0	1	1	Unsigned	1	0		0 = normal 1 = faulty
MOS high temperature limit for electrical control box	Intel	0	2	1	Unsigned	1	0		0 = normal 1 = faulty
MOS high temperature shutdown of electrical control box	Intel	0	3	1	Unsigned	1	0		0 = normal 1 = faulty
Electric control box headlight fault flag	Intel	1	8	1	Unsigned	1	0		0 = normal 1 = faulty
Electric control box brake light fault flag	Intel	1	9	1	Unsigned	1	0		0 = normal 1 = faulty
Fault indicator for the left turn signal of the electrical control box	Intel	1	10	1	Unsigned	1	0		0 = normal 1 = faulty
Fault indicator for the right turn signal in the electrical control box	Intel	1	11	1	Unsigned	1	0		0 = normal 1 = faulty

Vehicle fault status feedback									
Message Name			ID				Cycle (ms)		Message Length (Byte)
Veh_fb_Diag			0x18C4EAEF				10		8
Signal Description	Format	Starting Byte	Starting Bit	Signal Length	Data Type	Precision	Offset	Unit	Signal Value Description
Fault level	Intel	0	0	4	Unsigned	1	0		0: no fault 1: level 1 fault 2: level 2 fault 3: level 3 fault Other alarm invalid
Auto speed control CAN communication error	Intel	0	4	1	Unsigned	1	0		0 = normal 1 = failed
Auto IO control CAN communication error	Intel	0	5	1	Unsigned	1	0		0 = normal 1 = failed
Remote Driving Control CAN Communication Error	Intel	0	6	1	Unsigned	1	0		0 = normal 1 = failed
Auto torque control CAN communication error	Intel	0	7	1	Unsigned	1	0		0 = normal 1 = failed
EPS offline error	Intel	1	8	1	Unsigned	1	0		0 = normal 1 = failed
EPS error	Intel	1	9	1	Unsigned	1	0		0 = normal 1 = failed
EPS MOSFET over temperature	Intel	1	10	1	Unsigned	1	0		0 = normal 1 = failed
EPS alarm error	Intel	1	11	1	Unsigned	1	0		0 = normal 1 = failed
EPS working error	Intel	1	12	1	Unsigned	1	0		0 = normal 1 = failed
EPS over current error	Intel	1	13	1	Unsigned	1	0		0 = normal 1 = failed
EHB error feedback	Intel	2	20	12	Unsigned	1	0		0 = normal error status as per Note indicate
Driving motor controller error feedback	Intel	4	32	12	Unsigned	1	0		0 = normal error status as per Note indicate
BMS CAN communication offline error	Intel	5	44	1	Unsigned	1	0		0 = normal 1 = failed
E-stop error	Intel	5	45	1	Unsigned	1	0		0 = open 1 = close

Remote controller power off alarm	Intel	5	46	1	Unsigned	1	0	0 = normal 1 = failed
Remote controller receiver offline error	Intel	5	47	1	Unsigned	1	0	0 = normal 1 = failed
Alive Rolling Counter Heartbeat Signal (loop counter)	Intel	6	52	4	Unsigned	1	0	For each sent frame, the value will increase by 1, after the maximum value is reached, the value will be reset to 0 to check packet loss and disconnection.
Check BCC XOR checkout for message	Intel	7	56	8	Unsigned	1	0	Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6

Note: EHB error feedback explanation

Signal ID	Name	Description	Level
0x00		EHB No Fault	0
0x01	DiagEHB_DisOnlie	EHBOffline Fault	3
0x02	DiagEHB_CANErr	EHB CAN Communication Fault	3
0x03	DiagEHB_ECUUnderVolt	EHB ECU Undervoltage Fault	2
0x04	DiagEHB_ECUOverVolt	EHB ECU Overvoltage Fault	2
0x05	DiagEHB_ECUTempErr	EHB ECU Temperature Sensor Fault	2
0x06	DiagEHB_DomainDriveSCErr	EHB Domain Drive Circuit Short Fault	2
0x07	DiagEHB_DomainDriveOV	EHB Domain Drive Circuit Overvoltage Fault	2
0x08	DiagEHB_DomainDriveOT	EHB Domain Drive Circuit Overtemperature Fault	2
0x09	DiagEHB_AngleAlignmErr	EHB Angle Sensor Centering Fault	2
0x0A	DiagEHB_AngleSensorPOCErr	EHB Angle Sensor P Open Circuit	2
0x0B	DiagEHB_AngleSensorPAFErr	EHB Angle Sensor P Frequency Anomaly	2
0x0C	DiagEHB_AngleSensorSOCErr	EHB Angle Sensor S Open Circuit	2
0x0D	DiagEHB_AngleSensorSAFErr	EHB Angle Sensor S Frequency Anomaly	2
0x0E	DiagEHB_AngleOverErr	EHB Angle Sensor Fault	2
0x0F	DiagEHB_PreSensorErr	EHB Oil Pressure Sensor Fault	2
0x10	DiagEHB_InitOilErr	EHB Initial Oil Pressure Error	2
0x11	DiagEHB_PreSensprCIErr	EHB Oil Pressure Sensor Credibility Anomaly	2
0x12	DiagEHB_oilErr	EHB Oil Pressure Deviation Error	1

0x13	DiagEHB_MtrStallErr	EHB Motor Stall Error	2
0x14	DiagEHB_MtrDcErr	EHB Motor Disconnection Error	2
0x15	DiagEHB_MtrNoloadErr	EHB Motor Freewheeling	2
0x16	DiagEHB_SendValueErr	EHB VCU Command Reception Error	2
0x17	DiagEHB_EHBModelErr	EHB Working Mode Error	2
0x18	DiagEHB_disEnable	EHB Not Enabled Error	2
0x19	DiagEHB_AutoBrakeStErr	EHB Automatic Braking Mode Error	2
0x1A	DiagEHB_LowVoltCharging	EHB charging error under low voltage	2
0x1B	DiagEHB_EHBKeyFault	Electric control box EHB ignition fault	3

Note: Driving motor controller fault feedback signal explanation

Signal ID	Name	Description	Level
0x00		Motor Driver No Fault	0
0x01	DiagMCU_DisOnlie	Motor Driver CAN Communication Signal Lost	3
0x02	DiagMCU_UPhaseOverCurrent	Motor Driver U-Phase Overcurrent Fault	3
0x03	DiagMCU_VPhaseOverCurrent	Motor Driver V-Phase Overcurrent Fault	3
0x04	DiagMCU_WPhaseOverCurrent	Motor Driver W-Phase Overcurrent Fault	3
0x05	DiagMCU_HardwareOverCurrent	Motor Driver Hardware Overcurrent	3
0x06	DiagMCU_PowerModuleFault	Motor Driver Power Module Fault	3
0x07	DiagMCU_DCOverCurrent	Motor Driver Bus Overcurrent	3
0x08	DiagMCU_DCOverVolt	Motor Driver Bus Overvoltage	3
0x09	DiagMCU_DCUnderVolt	Motor Driver Bus Undervoltage	3
0x0A	DiagMCU_MotorOverRPM	Driving Motor Overspeed Fault	3
0x0B	DiagMCU_MotorOverLoad	Driving Motor Overload Fault	3
0x0C	DiagMCU_MCUOverLoad	Motor Driver Overload Fault	3
0x0D	DiagMCU_MotorOverTemp	Driving Motor Overtemperature Fault	3
0x0E	DiagMCU_MCUOverTemp	Motor Driver Overtemperature Fault	3
0x0F	DiagMCU_MotorTempSensFault	Driving Motor Temperature Sensor Fault	3
0x10	DiagMCU_MCUTempSensFault	Motor Driver Temperature Sensor Fault	3
0x11	DiagMCU_MotorEncoderFault	Motor Driver Motor Encoder Fault	3
0x12	DiagMCU_LockedRotorFault	Driving Motor Stall Fault	3
0x13	diagMCU_directionHPDFault	Driving Motor Direction HPD Error	3
0x14	diagMCU_drive2Fault	Motor Driver Drive 2 Fault	3
0x15	diagMCU_drive1Fault	Motor Driver Drive 1 Fault	3
0x16	DiagMCU_PhCurrentSensorFault	Motor Driver Phase Current Sensor Fault	3
0x17	DiagMCU_DCCurrentSensFault	Motor Driver Bus Current Sensor Fault	3
0x18	DiagMCU_DisCtrlFault	Driving Motor Runaway Fault	3

0x19	diagMCU_MotorHPDFault	Driving Motor HPD Error	3
0x1A	diagMCU_MOSFault	Motor Driver MOS Fault	3
0x1B	DiagMCU_PreChargeFault	Motor Driver Precharge Fault	3
0x1C	DiagMCU_CANFault	Motor Driver Communication Fault	3
0x1D	diagMCU_PreChangeOvertimeFault	Motor Driver Precharge Timeout Fault	3
0x1E	diagMCU_CapacitorShortFault	Motor Driver Capacitor Bank Short Circuit Fault	3
0x1F	diagMCU_EEPROMFault	Motor Driver EEPROM Fault	3
0x20	diagMCU_ConfDataFault	Motor Driver Configuration Parameter Fault	3
0x21	DiagMCU_LosePhaseFault	Motor Driver Phase Loss Fault	3
0x22	DiagMCU_PowerOnFault	Motor Driver Power-On Fault	3
0x23	DiagMCU_ParkOpenFault	Electromagnetic Brake Open Circuit Fault	3
0x24	DiagMCU_ParkOverCurrentFault	Electromagnetic Brake Overcurrent Fault	3
0x25	diagMCU_CurrentSensorErrLimt	Bus Current Sensor Anomaly Power Limi	2
0x26	diagMCU_DCCurrentLimt	Motor Driver Bus Current Power Limit	2
0x27	diagMCU_DCCurrentSensErr	Motor Driver Bus Current Sensor Anomaly Power Limit	2
0x28	diagMCU_EcoderErrLimt	Driving Motor Encoder Anomaly Power Limit	2
0x29	diagMCU_MCUOverTempLimt	Motor Driver Overtemperature Limit	2
0x2A	diagMCU_MotorOverCurrentLimt	Driving Motor Overcurrent Limit	2
0x2B	diagMCU_MotorOverTempLimt	Driving Motor Overtemperature Limit	2
0x2C	diagMCU_OverRPMLimt	Driving Motor Motor Overspeed Limit	2
0x2D	diagMCU_OverPowerLimt	Driving Motor Overpower Limit	2
0x2E	diagMCU_OverLoadLimt	Driving Motor Overload Limit	2
0x2F	dianMCU_UnVoltageLimt	Driving Motor Undervoltage Power Limit	2
0x30	DiagMCU_MCUKeyFault	Electric control box motor driver ignition fault	3

4.3.2. CAN cable connection

The CAN wires of FR-mid have all been soldered and labeled for easy connection by users according to the markings, as shown in Figure 4-1



Figure 4-1 CAN cable

4.3.3. VCU protocol instruction

1. Precautions During Testing:

- (1) During transmission, ensure the AliveCounter changes continuously in a cyclic manner.
- (2) Pay special attention to the fact that the AliveCounter occupies bits 52 to 55 (four bits).
- (3) The checksum in BYTE[7] is the XOR of the first 7 bytes:
Checksum = Byte0 XOR Byte1 XOR Byte2 XOR Byte3 XOR Byte4 XOR Byte5 XOR Byte6
- (4) The following routine is a simplified control command when sending via USB CAN.
When controlling the vehicle, ensure commands are sent according to the communication protocol.
- (5) During testing, switch the remote controller to autonomous mode or turn it off.
- (6) When testing with a CAN adapter connected to a PC, since the vehicle may move, lift the vehicle during testing. Only lower it after ensuring stability in the test program.

- (7) During ground testing, since the remote controller has the highest priority, it is best to keep it on for easy switching to remote control mode if needed.
- (8) The remote control has torque control mode disabled by default.
- (9) In remote torque control mode, push the throttle lever slowly following normal driving habits to prevent hazards caused by rapid acceleration due to excessive throttle opening or operator unfamiliarity with the control.
- (10) The priority of vehicle control commands is: Remote torque control command > Auto speed control command > Auto torque control command.

2. Control Command Description (ctrl_cmd)

Using the auto_spd_ctrl_cmd instructions as an example: when the corresponding command needs to be controlled, the command itself, the heartbeat signal, and the check digit must be sent simultaneously. If a CAN card host computer is used for control, the heartbeat signal must change; therefore, two or more CAN signal frames need to be sent cyclically during the control process.

(1) Target Gear Request (ctrl_cmd_gear)

The **ctrl_cmd_gear** command represents the target gear signal, with a physical value range of 01 to 04. The default gear position is 01 (Park - P).

03: Neutral (N)

02: Reverse (R)

04: Drive (D)

01: Park (P)

Example:

If the target gear request is Drive (D), the value is 04 (0x04).

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2D0	0x04	0x00	0x00	0x00	0x00	0x00	0x10	0x14
0x18C4D2D0	0x04	0x00	0x00	0x00	0x00	0x00	0x20	0x24
0x18C4D2D0	0x04	0x00	0x00	0x00	0x00	0x00	0x30	0x34

Note: The above three frames should be cyclically transmitted at 10ms intervals to enable gear shifting to D (Drive) mode.

Return:

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2EF	0x04	0x00	0x00	0x00	0x00	0x00	0x00	0x04

Note: The checksum and AliveCounter change cyclically.

(2) Target Speed Request (ctrl_cmd_velocity)

The **ctrl_cmd_velocity** command specifies the target speed value for vehicle propulsion, with a CAN communication physical range of 0 to 65.535 m/s (maximum speed of 6.6 m/s for a 10:1 speed ratio and 420mm wheel diameter). The target speed is determined by the speed resolution (0.001 m/s/bit).

Target vehicle speed = 0.001 * bus signal.

The vehicle's forward/reverse motion is coordinated with the gear position.

Speed feedback is provided in three modes:

- 1) **Current speed feedback:** Always returns a positive value.
- 2) **Left/right wheel speed feedback:** Indicates the respective speeds of the left and right wheels (positive for forward, negative for reverse).
- 3) **Left/right wheel pulse count feedback:** Pulse count increments during forward motion and decrements during reverse.

Example:

For a forward speed request of 5 m/s, the bus signal value is 5000 (0x1388).

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2D0	0x84	0x38	0x01	0x00	0x00	0x00	0x00	0xBD
0x18C4D2D0	0x84	0x38	0x01	0x00	0x00	0x00	0x10	0xAD
0x18C4D2D0	0x84	0x38	0x01	0x00	0x00	0x00	0x20	0x9D

Note: The above three frames should be transmitted cyclically at 10ms intervals to maintain the vehicle's forward motion at 5m/s.

Return:

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2EF	0x84	0x38	0x01	0x00	0x00	0x00	0x00	0xBD

Note: The checksum and AliveCounter change cyclically. Due to automatic speed adjustment during operation, the actual feedback speed may not be exactly 5m/s.

Left wheel speed and pulse feedback ID: 0x18C4D7EF

Right wheel speed and pulse feedback ID: 0x18C4D8EF

(3) Target Steering Angle (ctrl_cmd_steering)

The ctrl_cmd_steering command specifies the target steering angle request, with a CAN communication physical range of -40.96° to $+40.95^{\circ}$. The vehicle's internal software limits the steering angle to -32° to $+32^{\circ}$.

Left turn: Positive value

Right turn: Negative value

The target steering angle is determined by a resolution of $0.01^{\circ}/\text{bit}$:

Target steering angle = Bus signal $\times$ 0.01

Example:

For a target steering angle of -25° , the bus signal value is -2500 (0xF63C).

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2D0	0x00	0x00	0XC0	0x63	0x0F	0x00	0x00	0xAC
0x18C4D2D0	0x00	0x00	0XC0	0x63	0x0F	0x00	0x10	0xBC
0x18C4D2D0	0x00	0x00	0XC0	0x63	0x0F	0x00	0x20	0x8C

Note: The above three frames should be transmitted cyclically at 10ms intervals to set the steering angle request to -25 degrees.

Return:

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2EF	0x00	0x00	0XC0	0x63	0x0F	0x00	0x00	0xAC

Note: The checksum and AliveCounter change cyclically.

(4) Brake Request (ctrl_cmd_Brake)

The ctrl_cmd_Brake command specifies the vehicle's braking request percentage, with a CAN communication physical range of 0% to 100%:

- 0%: Brake fully released
- 100%: Maximum braking request

The feedback status also ranges from 0% to 100%, where:

- 0%: Brake completely released
- 100%: Maximum braking engagement

Example:

For a 100% braking request: 0x64

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2D0	0x00	0x00	0X00	0x00	0x40	0x06	0x00	0x46
0x18C4D2D0	0x00	0x00	0X00	0x00	0x40	0x06	0x10	0x56
0x18C4D2D0	0x00	0x00	0X00	0x00	0x40	0x06	0x20	0x66

Note: The above three frames should be transmitted cyclically at 10ms intervals to execute the braking request.

Return: When brake engagement is 100%:

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D2EF	0x00	0x00	0X00	0x00	0x40	0x06	0x00	0x46

Note: The checksum and AliveCounter change cyclically.

(5) Auxiliary Control Command Description

Taking the clearance lamp enable as an example (other auxiliary controls follow the same principle as clearance lamp enable control). The IO port enable control requires simultaneous transmission of the enable flag, heartbeat signal, and checksum. (If IO control is disabled, all lighting controls will be managed by the VCU.)

Example:

io_cmd_clearance_lamp - Clearance lamp enable control

0x01: Enable

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4D7D0	0x01	0x20	0X00	0x00	0x00	0x00	0x00	0x21
0x18C4D7D0	0x01	0x20	0X00	0x00	0x00	0x00	0x10	0x31
0x18C4D7D0	0x01	0x20	0X00	0x00	0x00	0x00	0x20	0x01

Note: The above three frames should be transmitted cyclically at 50ms intervals to activate the high beam.

Return:

ID	D[0]	D[1]	D[2]	D[3]	D[4]	D[5]	D[6]	D[7]
0x18C4DAEF	0x01	0x20	0X00	0x00	0x00	0x00	0x00	0x21

Note: The checksum and AliveCounter change cyclically.

The auxiliary enable control supports clearance lamp control and left/right turn signal control. The brake lights are not controlled by CAN signals but are entirely managed by the VCU, which provides feedback on their enable status.

5. Precautions

This section contains some important considerations for the use, storage, and development of the FR-mid.

5.1. Battery Usage Precautions

▲ The FR-mid product may not be fully charged when shipped. The remaining battery level can be checked via the FR-mid remote control's vehicle chassis display or through the CAN bus communication interface. Charging is complete when the charger's green indicator light turns on.

▲ Do not wait until the battery is completely depleted before recharging. Recharge promptly when the remaining battery level drops below 10%.

▲ The operating temperature range for battery discharge is -20°C to 50°C. The battery functions normally within this range; exceeding it may trigger battery protection.

▲ Charge the battery at temperatures above 0°C. Below 0°C, the battery may enter protection mode and refuse to charge.

▲ Avoid excessive discharge during use, as it may damage the battery.

▲ Prevent excessive impact on the battery, as impacts beyond specifications may cause leakage, overheating, smoking, fire, or explosion.

▲ If the battery shows obvious abnormalities, stop using it immediately!

5.2. Charging Precautions

▲ Only use the dedicated charger provided with the battery. Do not use non-standard batteries, power supplies, or chargers.

▲ Charge only in environments between 10°C and 45°C. Charging outside this range may cause battery leakage, overheating, severe damage, or deterioration in performance and lifespan.

▲ If the charger or battery behaves abnormally or is damaged during charging, unplug the charger's input and output power cables immediately.

▲ If charging cannot be completed within the specified time, stop the charging process. Otherwise, the battery may overheat, emit smoke, catch fire, or explode.

▲ Do not charge the vehicle battery during thunderstorms.

▲ Do not charge the vehicle battery in damp, rainy, or submerged locations.

▲ Do not charge the vehicle battery near heat sources or in direct sunlight where temperatures are excessively high.

▲ Charge in a well-ventilated, dust-free area.

▲ Do not block the charger's air vents during charging; maintain at least 10cm of clearance.

5.3. Usage Environment Precautions

▲ The FR-mid operates between -20°C and 50°C. Do not use it in temperatures below -20°C or above 50°C.

▲ The optimal storage temperature for the FR-mid is 0°C to 25°C.

▲ Do not store or use in environments with corrosive, flammable, or explosive gases.

▲ Keep away from heat sources and open flames during use and storage.

▲ Unless it is a specially customized version (with IP protection rating), the FR-mid has limited waterproofing. Avoid using it in deep water.

5.4. Remote Control Operation Precautions

▲ When debugging with the remote control, ensure it is powered on and that the

vehicle can receive commands. If the remote signal is weak, the remote will alarm—move closer to the vehicle.

▲ Before powering on, ensure all DIP switches are in the uppermost position, the emergency stop switch is released, and the throttle lever is at zero (chassis speed = 0).

▲ When using remote control, prioritize low-speed gear operation. Test medium or high speeds only after becoming familiar with the vehicle.

5.5. Electrical Expansion Precautions

▲ The top expansion power supply current must be used strictly according to the selected battery voltage and current specifications. Do not overload.

▲ When plugging or unplugging the top expansion power interface, turn off the vehicle switch to avoid short circuits.

▲ If the system detects that the battery voltage is below the safe threshold, it will activate protection. If external expansion devices involve critical data storage without automatic power-loss protection, recharge promptly.

5.6. Other Precautions

▲ Do not drop or invert the device during transportation or setup.

▲ Do not disassemble the device without professional authorization.

▲ If the remote control terminal is unused for an extended period, remove its batteries.

▲ Replace tires promptly if the tread pattern shows significant wear.

6. Common Q&A

Q : The FR-mid starts normally, but the vehicle does not move when controlled by the remote. What should I do?

A : First, check if the rear emergency stop switch is released. Then verify whether the SWA lever is in remote control mode, whether the VRB knob is unlocked, and whether the SWB gear lever matches the control command.

Q : What should I do if the FR-mid remote runs out of power and the vehicle stops?

A : Replace the remote control batteries immediately. Communication will resume once the remote is powered on again.

Q : Can the chassis shell be removed for vehicle body modifications?

A : This chassis uses a non-load-bearing body design, meaning the shell does not support internal components. Removing the shell has minimal impact, but it is not recommended. If necessary, have it done by professionals. The company is not responsible for damages caused by unauthorized disassembly.

Q : Neither LED1 nor LED2 on the charger lights up.

A : First, check if the charger's input cable is securely connected. Then verify whether

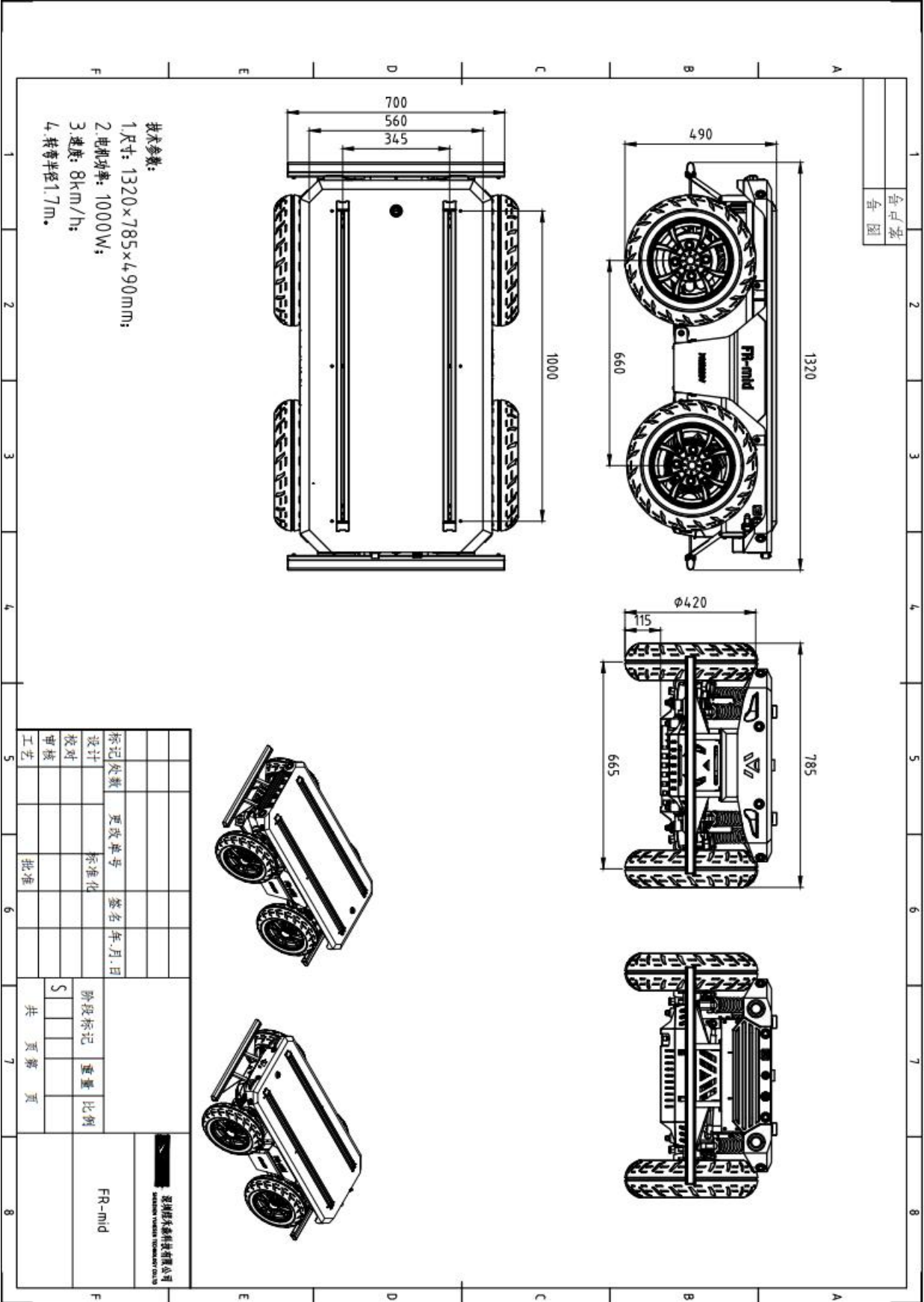
there is AC power input.

- Is the battery unused for a long time, over-discharged, or damaged?
- Replug the input and output cables with an interval of more than 10 seconds to check if the charger is in protection mode.

Q: How to turn off the remote control while the receiver remains powered?

A: Remove and reinsert the remote control batteries to shut down the remote while keeping the receiver powered.

7. Specification



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The logo for YUHESEN, featuring the company name in a bold, blue, italicized sans-serif font.

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